CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

3.1 Momentum & Newton's Laws of Motion

Syllabus Section: 3. Dynamics **Paper Type:** Multiple Choice (Paper 1) **Total:** 23 Questions (23 Marks)

Question 1 Easy [1 mark]

A tennis ball is dropped from different heights and its velocity is measured just before it hits the ground.

A graph is plotted showing the momentum of the tennis ball against its velocity.

What does the gradient of this graph represent?

A the acceleration of the tennis ball

B the mass of the tennis ball

C the kinetic energy of the tennis ball

D the velocity of the tennis ball

Mark Scheme & Explanation:

The correct answer is B

- The gradient of a momentum-velocity graph is equal to

$$\frac{\Delta y}{\Delta x} = \frac{\text{change in momentum}}{\text{change in velocity}} = \frac{\Delta p}{\Delta v}$$

- Momentum is equal to $p = mv$

$$\frac{\Delta p}{\Delta v} = \frac{\Delta (mv)}{\Delta v}$$

- For constant mass:

$$\frac{\Delta (mv)}{\Delta v} = m\frac{\Delta v}{\Delta v} = m$$

- Therefore, the gradient is equal to the mass of the tennis ball

Question 2 Easy [1 mark]

A cyclist is riding at a steady speed on a level road

According to Newton's third law of motion, what is equal and opposite to the backward push of the back wheel on the road?

A the force exerted by the cyclist on the pedals

B the total air resistance and friction force

C the tension in the cycle chain

D the forward push of the road on the back wheel

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The question tells you a key statement about Newton's Third Law (that the force is equal and opposite)
- But there are two key pieces of information about the forces in Newton's Third Law you will need to answer this question
- Firstly, the force acts between the same two objects. So, if one force is from the back wheel on the road, the other **must** be from the road on the back wheel
- Secondly, the two forces must be the same kind of force. In this particular case, the force on the wheel on the road is a friction force, so the force of the road on the wheel must be a friction force, too

A, B & C are incorrect because the force must act between the same two objects (the back wheel and the road), and be the same force (frictional) whereas these options all reference different bodies and different forces

Question 3 Easy [1 mark]

A resultant force causes a body to accelerate.

What is equal to the resultant force?

A the acceleration of the body per unit mass

B the change in kinetic energy of the body per unit time

C the change in momentum of the body per unit time

D the change in velocity of the body per unit time

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The resultant force is equal to the rate of change of momentum, which is the same as saying the change in momentum of the body per unit time
- This is given by Newton's second law:
  - $F = ma = \frac{m\Delta v}{t} = \frac{\Delta p}{t}$

A is incorrect because 'acceleration per unit mass' is equivalent to $\frac{a}{m}$. According to Newton's second law, force is the product of mass and acceleration $F = ma$.

B is incorrect because a change in kinetic energy per unit time (rate of change of energy) is equivalent to power $P = \frac{E}{t}$.

D is incorrect because a 'change in velocity per unit time' (rate of change of velocity) is equivalent to acceleration $a = \frac{\Delta v}{\Delta t}$.

Question 4 Easy [1 mark]

What describes the mass and the weight of an object?

	Mass	Weight
A	the ratio of the object's momentum to its velocity	the work done in lifting the object 1.0 m vertically
B	the force the object experiences due to gravity	the resistance of the object to changes in motion
C	the resistance of the object to changes in motion	the force the object experiences due to gravity
D	the total number of atoms in the object	the ratio of the object's mass to the acceleration of free fall

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- **Mass** is the property of an object that resists a change in motion
  - It is a constant quantity and does not change with location
- **Weight** is the force exerted on an object due to a gravitational field
  - The weight of the object depends on its mass and the strength of the gravitational field at that location

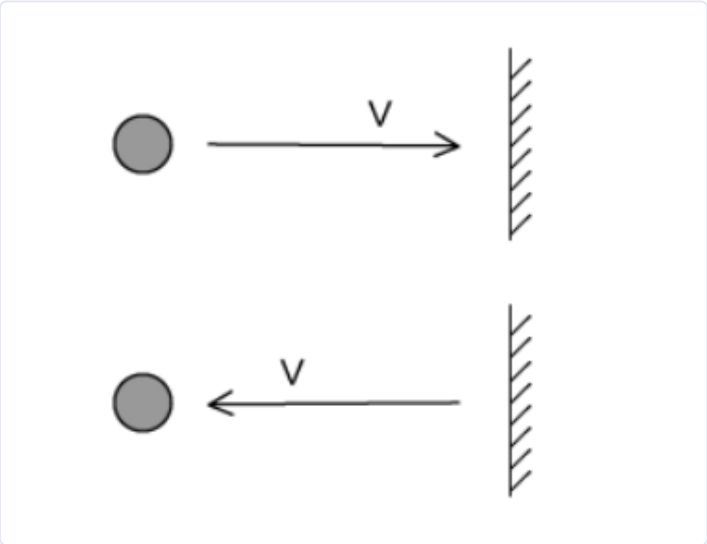
A is incorrect because while the work done lifting an object 1.0 m vertically is numerically related to weight $W = mgh$, it describes energy, whereas weight is a force.

B is incorrect because these definitions are correct, but are the wrong way around.

D is incorrect because an object's mass is related to the amount of matter it contains rather than the number of atoms.

Question 5 Easy [1 mark]

An object travelling with velocity *v* strikes a wall and rebounds as shown



Which property of the object is **not** conserved?

A momentum

B mass

C kinetic energy

D speed

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Momentum is a vector quantity
- Initial momentum = *mv*, final momentum = – *mv*
- Even though the magnitude of the momentum remains the same, the direction of the velocity changes
- The momentum of the object is not conserved but the momentum of the **whole system** should be conserved. The question asks which 'property of the object' is not conserved, not the whole system, so this is an important distinction

B, C & D are incorrect because these are scalar quantities, and the magnitudes of *m* and *v* stay the same so kinetic energy, speed and mass are all conserved

Question 6 Easy [1 mark]

A force *F* is applied to a freely moving object. At one instant of time, the object has velocity *v* and acceleration *a*.

Which quantities must be in the same direction?

A *a* and *v* only

B *v*, *a* and *F*

C *v* and *F* only

D *F* and *a* only

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- According to Newton's Second Law, a force causes an object to accelerate in the same direction it is applied

A, B & C are incorrect because Newton's First Law states that an object will remain at rest or at the same constant speed unless a resultant force acts upon it. This means a force could be applied in the same direction as an object's motion, causing it to accelerate, or an opposing force could be applied, to cause it to decelerate. Therefore, the direction of the velocity does not have to be the same direction as the force or acceleration

Question 7 Easy [1 mark]

Which is **not** one of Newton's laws of motion?

A if body A exerts a force on body B, then body B exerts an equal and oppositely-directed force on body A

B the rate of change of momentum of a body is directly proportional to the external force acting on the body and takes place in the direction of the force

C the total momentum of a system of interacting bodies remains constant, providing no external force acts

D a body continues in a state of rest or of uniform motion in a straight line unless acted upon by some external force

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- This is the principle of conservation of momentum which is a direct consequence of Newton's Third Law, but it does not describe it
- Newton's Third Law states that for every force there is an equal and opposite force, from this we can derive conservation of momentum, which tells us that a change in momentum in one of two objects colliding is equal and opposite to the change in momentum in the other colliding object

A is incorrect because this is Newton's Third Law

B is incorrect because this is Newton's Second Law, $F = ma = \frac{m\Delta v}{t} = \frac{\Delta p}{t}$

D is incorrect because this is Newton's First Law

Question 8 Easy [1 mark]

The acceleration of free fall is represented by g.

Which statement is correct?

A g is the force due to gravity

B g is reduced by air resistance

C g is the weight of an object

D g is the ratio of an object's weight to its mass

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Weight = mass × acceleration due to gravity, or $W = mg$
- So, $g = \frac{W}{m}$

A & C are incorrect because gravity/weight is a force, whereas g is the acceleration

B is incorrect because while air resistance reduces the acceleration of a falling body, it does not change the value of g itself.

Question 9 Easy [1 mark]

As a car accelerates uniformly, its momentum is measured at regular intervals, and a graph of the car's momentum is plotted against time.

Which quantity is equal to the gradient of the graph at a particular time?

A the acceleration of the car

B the resultant force on the car

C the kinetic energy of the car

D the velocity of the car

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The gradient of a momentum-time graph is equal to
  - $\frac{\Delta y}{\Delta x} = \frac{\Delta p}{\Delta t}$
- Momentum is equal to $\Delta p = m\Delta v$
  - $\frac{\Delta p}{\Delta t} = \frac{m\Delta v}{\Delta t}$
- Acceleration is equal to $a = \frac{\Delta v}{\Delta t}$
  - $m\frac{\Delta v}{\Delta t} = ma$
- From Newton's second law, $F = ma$
- Therefore, the gradient is equal to the resultant force on the car

Question 10 Medium [1 mark]

All external forces on a body cancel out.

Which statement **must** be correct?

A the body does not move

B the total energy (kinetic and potential) of the body remains unchanged

C the speed of the body is zero

D the momentum of the body remains unchanged

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- If all the external forces cancel out, that means the resultant force on the body is zero
- If the resultant force is zero then the acceleration is also zero
  - $F = ma$
- Zero acceleration means that the velocity is not changing
  - $a = \frac{v}{t}$
- Hence, the momentum of the body remains unchanged
  - $p = mv$

A is incorrect because if a body is not moving then its velocity is zero, however, if acceleration is zero then that does not necessarily mean the velocity is zero. The body may be moving at a constant velocity, but its acceleration would still be zero

B is incorrect because force and energy are not equivalent, so if the resultant force is zero this doesn't mean the energy will be unchanged

C is incorrect because, for a stationary body in equilibrium, its velocity must remain constant but its speed can be either zero or a constant non-zero value

Question 11 Medium [1 mark]

A ball falls vertically and bounces on the ground.

Which statement about the forces acting while the ball is in contact with the ground is correct?

A the force that the ball exerts on the ground is always equal to the weight of the ball

B the force that the ball exerts on the ground is always less than the weight of the ball

C the force that the ball exerts on the ground is always equal in magnitude and opposite in direction to the force the ground exerts on the ball

D the weight of the ball is always equal in magnitude and opposite in direction to the force that the ground exerts on the ball

Mark Scheme & Explanation:

*The correct answer is **C because**:*

- When the ball hits the ground, it rapidly slows down and then speeds up again as it bounces
- This is due to a changing contact force exerted by the ground on the ball
- According to Newton's third law:
    - When two bodies interact, they exert forces on each other that are equal in magnitude and opposite in direction
- Therefore, the ball must exert a force on the ground which is equal in magnitude and opposite in direction to the force the ground exerts on the ball

A is incorrect as the downward force that the ball exerts on the ground is the third law force pair of the upward contact force (i.e. it exists because of it). Since this contact force changes during the collision (as the ball decelerates and then accelerates upwards), the downward force must also change. Therefore, it is not always equal to the ball's weight.

B is incorrect because to accelerate the ball upwards, the ground must exert an upward contact force that is greater than the ball's weight. By Newton's third law, the downward force the ball exerts on the ground must also be greater than its weight. Therefore, it is not always less than the ball's weight.

D is incorrect because the weight of the ball is constant and always acts downwards (towards the ground), but the contact force from the ground (and therefore the force on the ball) changes during the collision.

Question 12 Medium [1 mark]

The acceleration of free fall on planet X is $\frac{1}{6}$ of the acceleration of free fall on Earth.

The mass of a body on planet X is 60 kg.

What is its weight on planet X?

A 10 N

B 59 N

C 98 N

D 360 N

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Mass is a constant quantity which does not change with a change of position or location, so the mass is 60 kg on planet X and on Earth
- The weight, however, changes depending on the strength of the gravitational field of a planet. This can be found using
  - $W = mg$
- Where $g = 9.81 \quad \text{m} \quad \text{s}^{-2}$ on Earth and $g = \frac{9.81}{6} = 1.63 \quad \text{m} \quad \text{s}^{-2}$ on planet X
- The weight of the body on planet X is
  - $W = \frac{60 \times 9.81}{6} = 98 \quad N$

Question 13 Medium [1 mark]

Newton's third law of motion is often summarised as

'Every action has an equal and opposite reaction.'

A cup of tea rests on a table. If the weight of the cup is the 'action' force, what is the 'reaction' force?

A the push of the table on the cup

B the push of the cup on the table

C the pull of the Earth on the cup

D the pull of the cup on the Earth

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Newton's Third Law says: when body A exerts a force on body B, body B exerts an equal and opposite force on body A of the **same type**
- For gravity, the Earth attracts the cup and the cup attracts the Earth with the same magnitude but opposite direction. Both forces are of the same type – gravitational attraction
- Consider a falling object – it is not just the object that is falling towards the earth but rather the earth too is falling towards the object. Given the huge difference in mass, the Earth's movement towards the object is so negligible it can be ignored

A & B are incorrect because the push of the table on the cup (and vice versa) is a result of the repulsive forces between the electrons on the surface of the table and the electrons on the surface of the bottom of the cup, both of these are electrostatic forces

C is incorrect because the pull of the Earth on the cup is the same as the weight

Question 14 Medium [1 mark]

The gravitational field strength on the surface of planet R is one eighth of that on the surface of planet S.

On the surface of R, a body has its mass measured to be 2.5 kg and its weight measured to be 12.5 N

What results are obtained for measurements of the mass and weight of the same body on the surface of planet S?

	Mass on S	Weight on S
A	2.5 kg	20 N
B	20 kg	100 N
C	2.5 kg	100 N
D	20 kg	160 N

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The mass of a body is constant and does not change, so it will be 2.5 kg on both planets
- The weight, however, changes depending on the strength of the gravitational field of a planet. This can be found using
  - $W = mg$
- Where g on planet S is 8 times greater than on planet R
- So, the weight of the body on planet S is
  - $W = 12.5 \times 8 = 100 \quad \text{N}$

A is incorrect because the mass is correct but the weight has been calculated using 2.5 × 8 = 20 which is wrong

B & D are incorrect because the mass should be unchanged

Question 15 Medium [1 mark]

Water is pumped through a hose-pipe at a rate of 80 kg per minute. It emerges from the hose-pipe horizontally with a speed of $30 \quad m \quad s^{-1}$

Which force is required from a person holding the hose-pipe to prevent it moving backwards?

A 3 N

B 40 N

C 160 N

D 2400 N

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The water that emerges from the hose has momentum, so it exerts a force on the person as it shoots out
- From Newton's Third Law, the person should exert a force of the same magnitude and in the opposite direction to prevent the hose from moving backwards
- Using Newton's Second Law
  - $F = ma = \frac{m\Delta v}{t} = \frac{\Delta p}{t}$ (force = rate of change of momentum)
  - $\frac{\Delta m}{t}$ = rate of water being pumped = 80 kg / min
- In one second, $\frac{80}{60} = 1.33 \quad \text{kg}$
- So, 1.3 kg of water is pumped out every second
  - $\frac{\Delta m}{t}$ = rate of water being pumped = $1.33 \quad \text{kg} \quad \text{s}^{-1}$
  - $F = \frac{\Delta p}{t} = \frac{\Delta m}{t} \times v = 1.33 \times 30 = 40 \quad \text{N}$

Question 16 Medium [1 mark]

A car is travelling at constant velocity on a straight road.

Which row gives the correct Newton's third law force pair?

	Force A	Force B
A.	Driving force from engine	Air resistance on the car
B.	Weight of the car	Normal reaction from the road
C.	Driving force from wheels on road	Friction force from air on car
D.	Weight of the car	Gravitational force of the car on the Earth

A

B

C

D

Mark Scheme & Explanation:

The correct answer is D

- Newton's third law states that whenever two bodies interact, the forces they exert on each other are equal and opposite and of the same type
- In other words, if object A exerts a force on object B, then object B exerts an **equal** but **opposite** force on object A
- Therefore, the forces must:
  - act on **different** objects
  - be the same **type** of force
  - be **opposite** in direction
  - be **equal** in magnitude
- The only force pair that satisfies all these conditions is option **D**

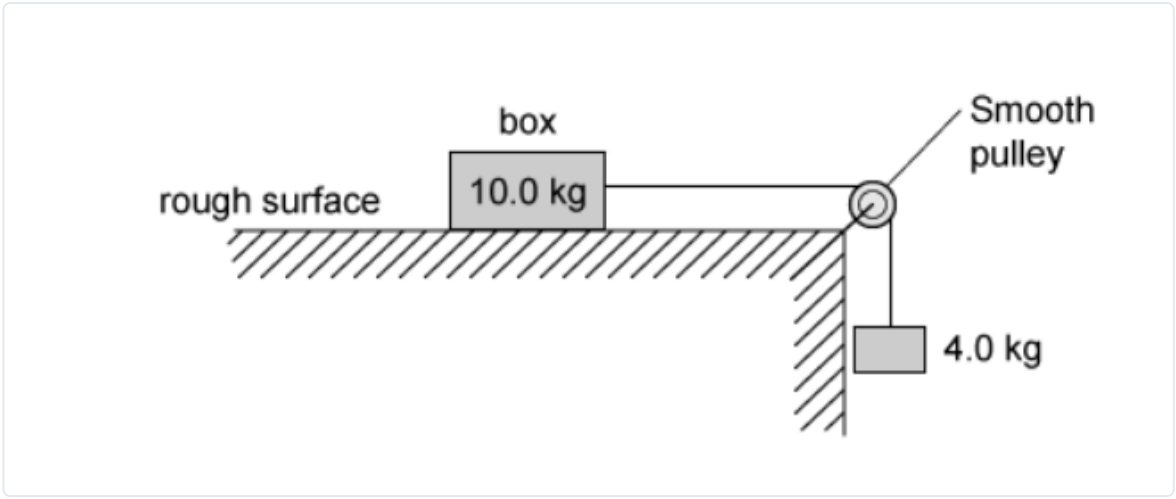
A is incorrect, as the driving force from the engine and air resistance are different types of forces (driving force is mechanical and air resistance is a fluid friction force). They also act on the same object (the car), so they cannot be a Newton's third law pair.

B is incorrect, as weight and normal reaction are two different types of forces (weight is gravitational and normal reaction is a contact force), and they also act on the same body (the car). Therefore, they cannot be a Newton's third law force pair. This is a common misconception of this law.

C is incorrect as these forces act on the same object (the wheels/car system) and are both contact forces in the same direction during acceleration, so they do not form a Newton's third law pair.

Question 17 Hard [1 mark]

A box of mass 10.0 kg rests on a horizontal rough surface. A string attached to the box passes over a smooth pulley and supports a 4.0 kg mass at its other end.



When the box is released, a frictional force of 12.0 N acts on it.

What is the acceleration of the box?

A $1.2\ \text{m}\ \text{s}^{-2}$

B $1.9\ \text{m}\ \text{s}^{-2}$

C $2.7\ \text{m}\ \text{s}^{-2}$

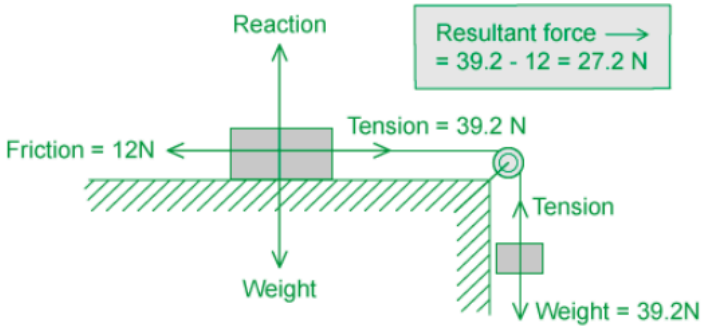
D $3.0\ \text{m}\ \text{s}^{-2}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

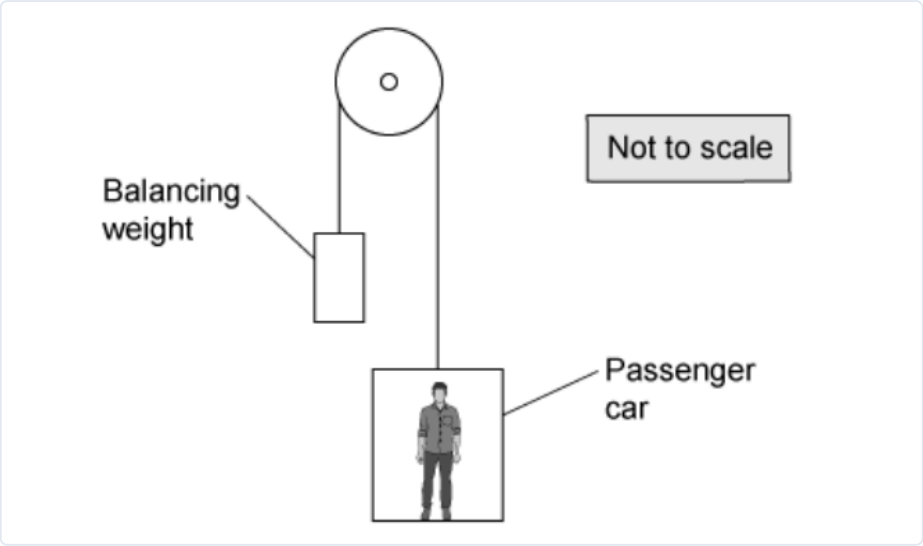
- The weight of the 4.0 kg mass pulls the 10.0 kg box to the right with a force equal to its weight
  - Weight = mg = 4.0 × 9.8 = 39.2 N
- The frictional force of 12.0 N opposes the motion of the 10.0 kg box, so it acts to the left
  - The resultant force is thus equal to 39.2 – 12.0 = 27.2 N
- It is important to remember that the mass and the box move as a system, so their masses must be added together
- Total mass of the system = 10.0 + 4.0 = 14.0 kg

- Resultant force, $F = ma$
- $a = \frac{F}{m} = \frac{27.2}{14} = 1.9 \text{ m s}^{-2}$



Question 18 Hard [1 mark]

A lift consists of a passenger car supported by a cable which runs over a light, frictionless pulley to a balancing weight. The balancing weight falls as the passenger car rises.



Some masses are shown in the table.

	Mass / kg
passenger car	1200
balancing weight	1320
passenger	80

What is the magnitude of the acceleration of the car when carrying just one passenger and when the pulley is free to rotate?

A $0.15\ m\ s^{-2}$

B $0.30\ m\ s^{-2}$

C $0.45\ m\ s^{-2}$

D $0.92\ m\ s^{-2}$

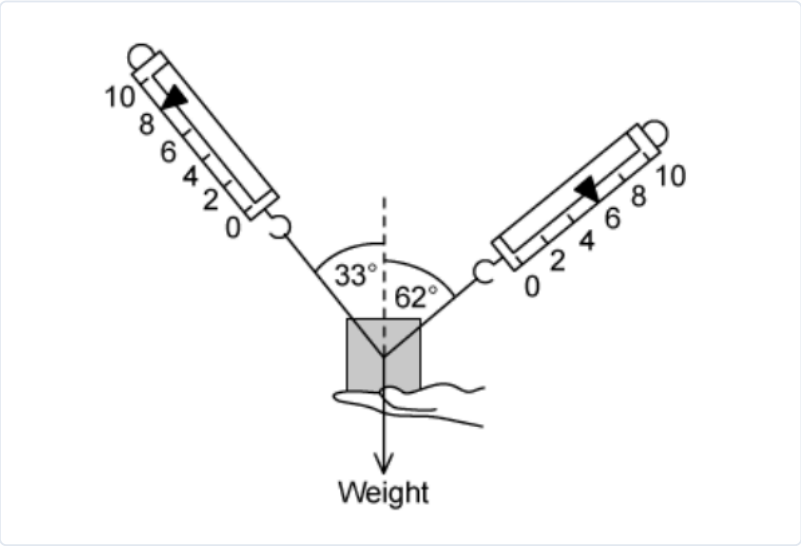
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The weight of the car and passenger causes the lift to move downwards, while the weight of the balancing weight causes the lift to move upwards
    - Resultant (upward) force on car, $F = 1320\ g - 1200\ g - 80\ g = 40\ g$
        - where $g = 9.81\ \text{N kg}^{-1}$
    - $F = 40 \times 9.81 = 392\ \text{N}$
- This resultant force is responsible for the acceleration of the passenger car, the balancing weight and the passenger, so we must use the total mass of the system, not just the mass of the lift and passenger
    - Total mass, $m = 1320 + 1200 + 80 = 2600\ \text{kg}$
- $F = ma$ rearranges to
    - $a = \frac{F}{m} = \frac{392}{2600} = 0.15\ \text{m s}^{-2}$

Question 19 Hard [1 mark]

A 3.2 kg mass is supported by a person's hand and two newton-meters as shown.



When the person's hand is removed, what is the initial vertical acceleration of the mass?

A $3.0\ \text{m s}^{-2}$

B $5.4\ \text{m s}^{-2}$

C $6.8\ \text{m s}^{-2}$

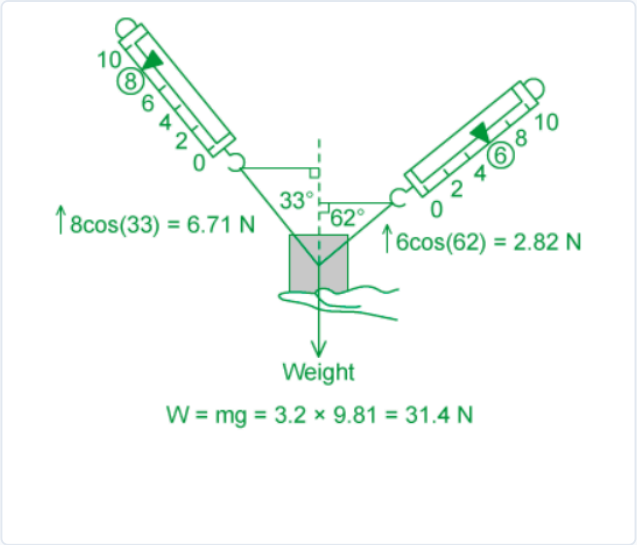
D $12.8\ \text{m s}^{-2}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

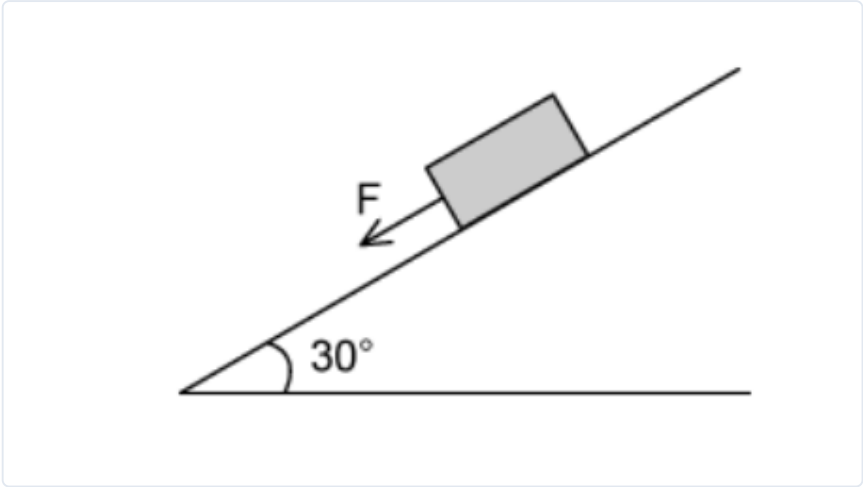
- We are looking for the vertical acceleration, so the horizontal components of force can be ignored
- The resultant upward force is equal to
  - $8 \cos 33° + 6 \cos 62° = 6.71 + 2.82 = 9.53\ \text{N}$
- The resultant downward force is equal to the weight of the block
  - W = mg = 3.2 × 9.81 = 31.4 N
- The overall resultant force is equal to
  - F = 31.4 – 9.53 = 21.9 N

- Acceleration = $\frac{F}{m} = \frac{21.9}{3.2} = 6.8 \text{ m s}^{-2}$



Question 20 Hard [1 mark]

A brick of mass 1.8 kg rests on a plane inclined at an angle of 30°. The weight of the brick has a component of *F* N parallel with the plane. The brick also experiences a frictional force of 3.0 N



What is the acceleration of the brick down the plane?

A 3.2 m s^{-2}

B 3.9 m s^{-2}

C 6.6 m s^{-2}

D 8.1 m s^{-2}

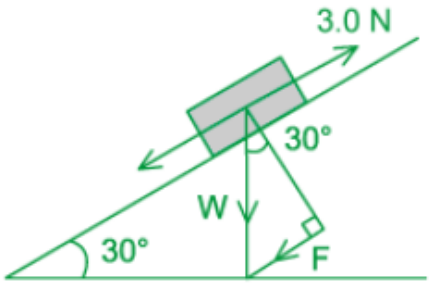
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Solving problems on a plane is similar to solving problems vertically and horizontally, but instead, we resolve the forces parallel and perpendicular to the plane
- Since we are trying to find the acceleration of the block, we need to resolve the forces parallel to the plane
  - Weight = *mg* = 1.8 × 9.8 = 17.6 N
  - Resultant force (down the plane) = W sin 30 = 17.6 sin 30 = 8.8 N
- The frictional force opposes the motion of the brick, so it acts up the plane
- So, the resultant force on the brick = 8.8 – 3.0 = 5.8 N
  - Acceleration of the brick, $a = \frac{F}{m} = \frac{5.8}{1.8} = 3.2$ m s^{-2}

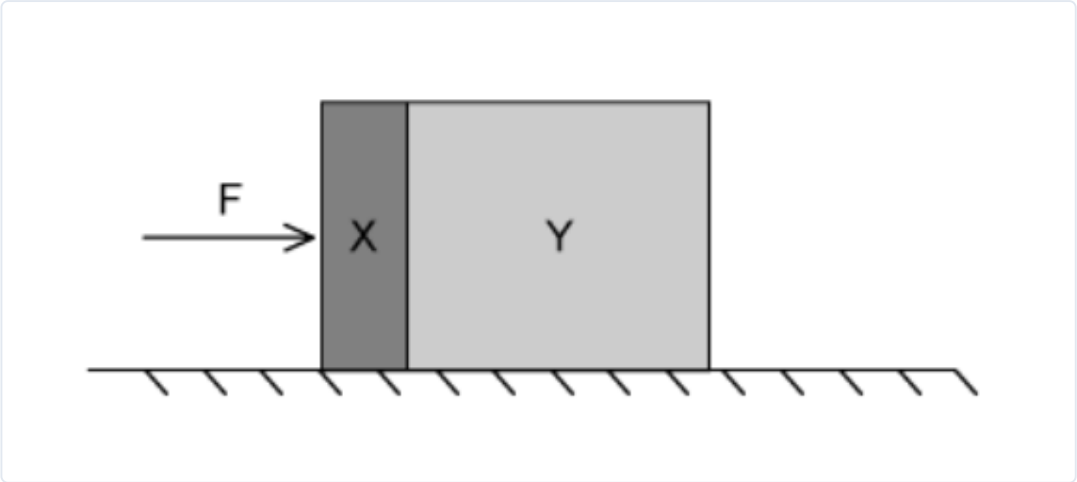
$W = 1.8 \times 9.8$
$= 17.6\ N$

$F = 17.6 \sin(30)$
$= 8.8\ N$



Question 21 Hard [1 mark]

Two blocks X and Y, of masses *m* and 4 *m* respectively, are accelerated along a smooth horizontal surface by a force *F* applied to block X as shown



What is the magnitude of the force exerted on block Y by block X during this acceleration?

A $\frac{F}{4}$

B $\frac{F}{5}$

C $\frac{4F}{5}$

D $\frac{5F}{4}$

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- The force *F* is applied to block X, which in turn applies a force to block Y
- As a result, both blocks accelerate
- Total mass being accelerated = *m* + 4 *m* = 5 *m*
- The acceleration of the blocks is therefore equal to
  - $a = \frac{F}{5m}$
- We can use Newton's Second Law again to calculate the force on block Y only
  - $F_y = 4m \quad a = 4m\frac{F}{5m} = \frac{4F}{5}$

Question 22 Hard [1 mark]

A ship of mass 6.7×10^7 kg is approaching a harbour with speed 18.9 m s^{-1}

By using reverse thrust it can maintain a constant total stopping force of 880,000 N

How long will it take to stop?

A 14 seconds

B 140 seconds

C 24 minutes

D 240 minutes

Mark Scheme & Explanation:

*The correct answer is **C** because:*

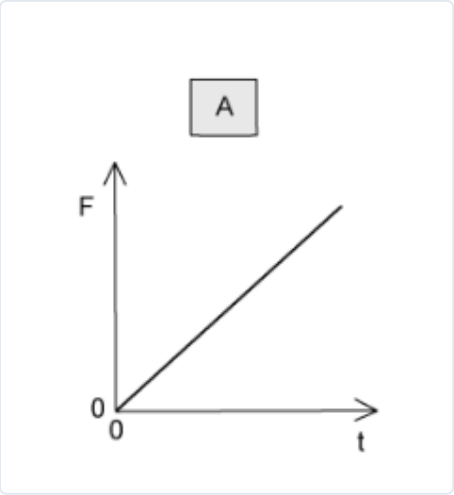
- Momentum is equal to
  - $\Delta p = m\Delta v$
- Force is related to momentum by
  - $F = \frac{\Delta p}{\Delta t}$
- Combining these leads to
  - $F = \frac{m\Delta v}{\Delta t}$
- Rearranging for Δt
  - $\Delta t = \frac{m\Delta v}{F}$
  - $\Delta t = \frac{6.7 \times 10^7 18.9 - 0}{880000}$
  - $\Delta t = 1439 \text{ s}$
- Divide by 60 to convert to minutes
  - $\Delta t = \frac{1439}{60} = 24 \text{ minutes}$

Question 23 Hard [1 mark]

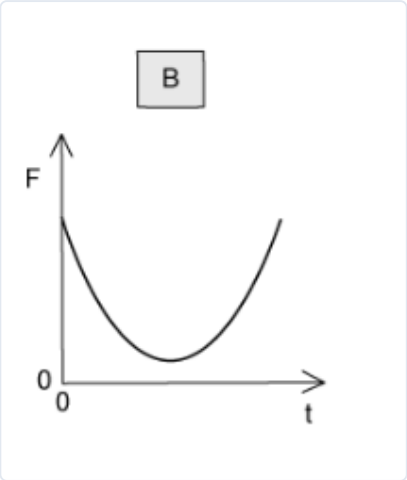
A tennis ball is dropped onto a table and bounces back up. The table exerts a force *F* on the ball.

Which graph best shows the variation with time *t* of the force *F* while the ball is in contact with the table?

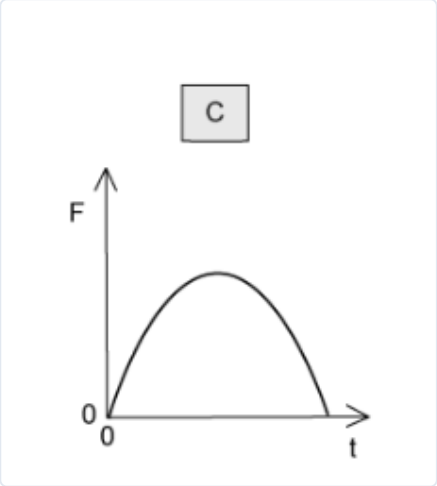
A



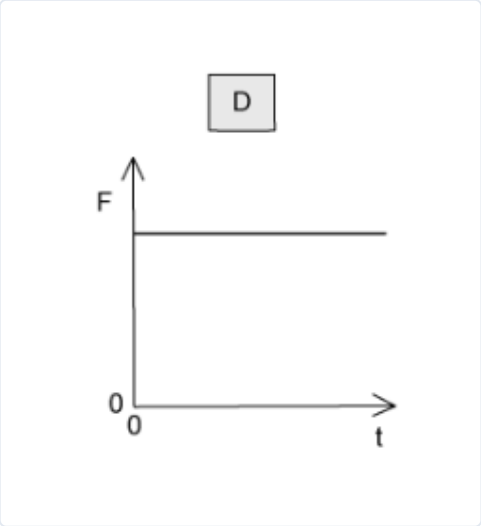
B



C



D



Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Force is equal to the rate of change of momentum, and therefore, it is directly proportional to the rate of change of velocity $F \propto \Delta v/\Delta t$
- Before the ball hits the table:
  - There is no force between the ball and the table, so the force is initially zero
  - Therefore, options **B** & **D** can be eliminated
- When the ball comes into contact with the table:
  - The ball's speed temporarily reduces to zero, this means there is a change in momentum, so F increases, which is shown by the initial positive gradient on the graph
  - When the speed is zero, it is no longer changing, so the change in momentum, and hence force, reaches a maximum, which is represented on the graph by the maximum point on the curve

    - Then, the speed of the ball increases in the opposite direction as it bounces away from the table (but the force acts only while the ball is still in contact with the table)
    - This increase in speed causes a change in momentum, and a force, but in the opposite direction, so as the ball moves away from the table, the force exerted by it gradually decreases, as seen on graph C
- At the moment the ball is about to leave the surface of the table, it has the same maximum speed as before and since the velocity/momentum is no longer changing, the force reduces to zero

A is incorrect because this graph implies that the force continues to increase, even after the ball leaves the table, which is not possible

B & D are incorrect because there is no force between the ball and the table before the ball comes into contact with it, hence the force is initially zero